In [ ]:
```
%pip install scikit-learn statsmodels seaborn scipy
```

In [ ]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from scipy import stats
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import (
    accuracy_score,
    confusion_matrix,
    classification_report,
    roc_auc_score
)
from sklearn.pipeline import Pipeline
from sklearn.preprocessing import StandardScaler
import statsmodels.api as sm
from statsmodels.stats.outliers_influence import variance_inflation_factor

print("Libraries imported successfully!")
```

In [8]:
```
df = pd.read_csv("train.csv")

print("Dataset loaded successfully!")
print("Dataset shape:", df.shape)

df.head()
```

```
Dataset loaded successfully!
Dataset shape: (891, 12)
```

Out[8]:

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	F
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.0

In [11]:

```
print("Missing values:")
print(df.isnull().sum())

print("\nData types:")
print(df.dtypes)
```

```
Missing values:
PassengerId      0
Survived         0
Pclass           0
Name             0
Sex              0
Age            177
SibSp            0
Parch            0
Ticket           0
Fare             0
Cabin          687
Embarked         2
dtype: int64

Data types:
PassengerId      int64
Survived         int64
Pclass           int64
Name               str
Sex                str
Age            float64
SibSp            int64
Parch            int64
Ticket             str
Fare           float64
Cabin              str
Embarked           str
dtype: object
```

In [13]:
```python
from scipy.stats import chi2_contingency

table = pd.crosstab(df["Sex"], df["Survived"])

chi2, p_value, dof, expected = chi2_contingency(table)

print("Chi-square statistic:", round(chi2, 3))
print("P-value:", p_value)

if p_value < 0.05:
    print("Reject H0: Gender and survival are significantly associated.")
else:
    print("Fail to reject H0: Evidence of association is insufficient.")
```

```
Chi-square statistic: 260.717
P-value: 1.1973570627755645e-58
Reject H0: Gender and survival are significantly associated.
```

In [15]:
```python
# Select relevant columns
model_df = df[
    ["Survived", "Pclass", "Sex", "Age", "Fare", "SibSp", "Parch"]
].copy()

# Fill missing values
model_df["Age"] = model_df["Age"].fillna(model_df["Age"].median())

# Convert gender into numeric values
model_df["Sex"] = model_df["Sex"].map({"male": 0, "female": 1})

# Check for missing values
```

```
print("Missing values after cleaning:")
print(model_df.isnull().sum())

print("\nPrepared dataset shape:", model_df.shape)
model_df.head()
```

```
Missing values after cleaning:
Survived    0
Pclass      0
Sex         0
Age         0
Fare        0
SibSp       0
Parch       0
dtype: int64

Prepared dataset shape: (891, 7)
```

Out[15]:

	Survived	Pclass	Sex	Age	Fare	SibSp	Parch
0	0	3	0	22.0	7.2500	1	0
1	1	1	1	38.0	71.2833	1	0
2	1	3	1	26.0	7.9250	0	0
3	1	1	1	35.0	53.1000	1	0
4	0	3	0	35.0	8.0500	0	0

In [16]:

```
X = model_df[["Pclass", "Sex", "Age", "Fare", "SibSp", "Parch"]]
y = model_df["Survived"]

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42, stratify=y
)

model = Pipeline([
    ("scaler", StandardScaler()),
    ("logistic", LogisticRegression(max_iter=1000))
])

model.fit(X_train, y_train)

y_pred = model.predict(X_test)
y_prob = model.predict_proba(X_test)[:, 1]

print("Accuracy:", round(accuracy_score(y_test, y_pred), 3))
print("ROC-AUC:", round(roc_auc_score(y_test, y_prob), 3))
print("\nConfusion Matrix:")
print(confusion_matrix(y_test, y_pred))
print("\nClassification Report:")
print(classification_report(y_test, y_pred))
```

```
Accuracy: 0.804
ROC-AUC: 0.852

Confusion Matrix:
[[95 15]
 [20 49]]

Classification Report:
              precision    recall  f1-score   support

           0       0.83      0.86      0.84       110
           1       0.77      0.71      0.74        69

    accuracy                           0.80       179
   macro avg       0.80      0.79      0.79       179
weighted avg       0.80      0.80      0.80       179
```

In [17]:
```python
# Logistic regression coefficients
logistic_model = model.named_steps["logistic"]

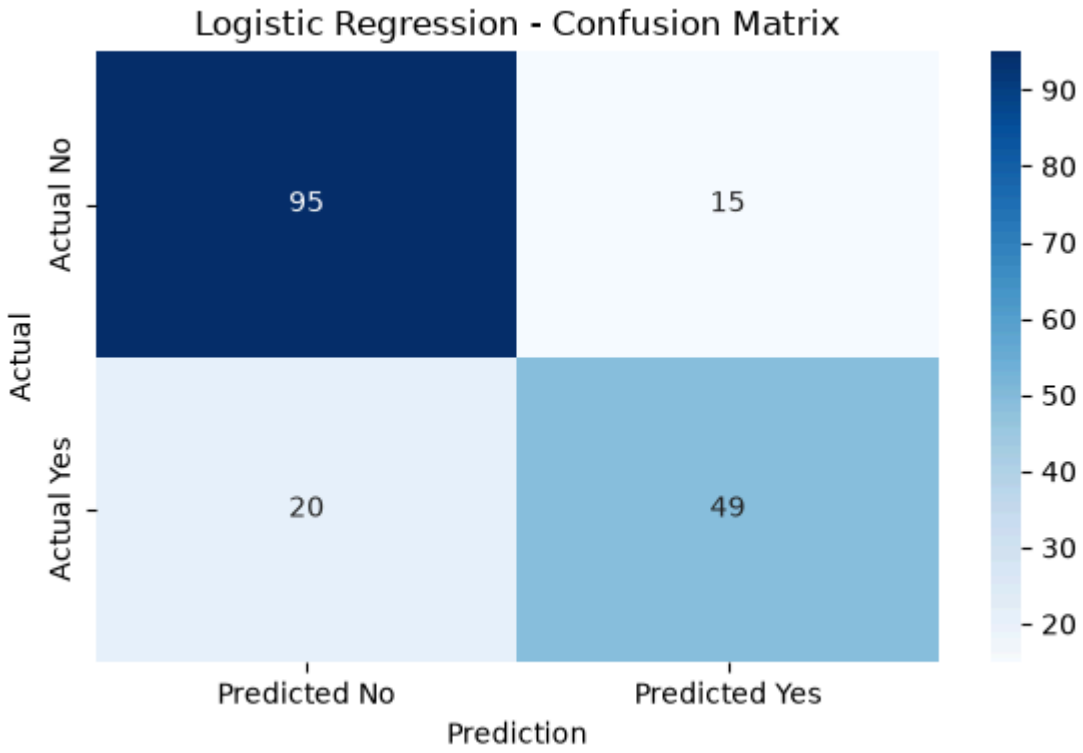
coef_table = pd.DataFrame({
    "Feature": X.columns,
    "Coefficient": logistic_model.coef_[0]
})

print("Logistic Regression Coefficients:")
print(coef_table.sort_values(
    by="Coefficient", ascending=False
).round(4))
```

```
Logistic Regression Coefficients:
  Feature  Coefficient
1     Sex       1.2969
3    Fare       0.1464
5   Parch      -0.0936
4   SibSp      -0.2868
2     Age      -0.5014
0  Pclass      -0.8936
```

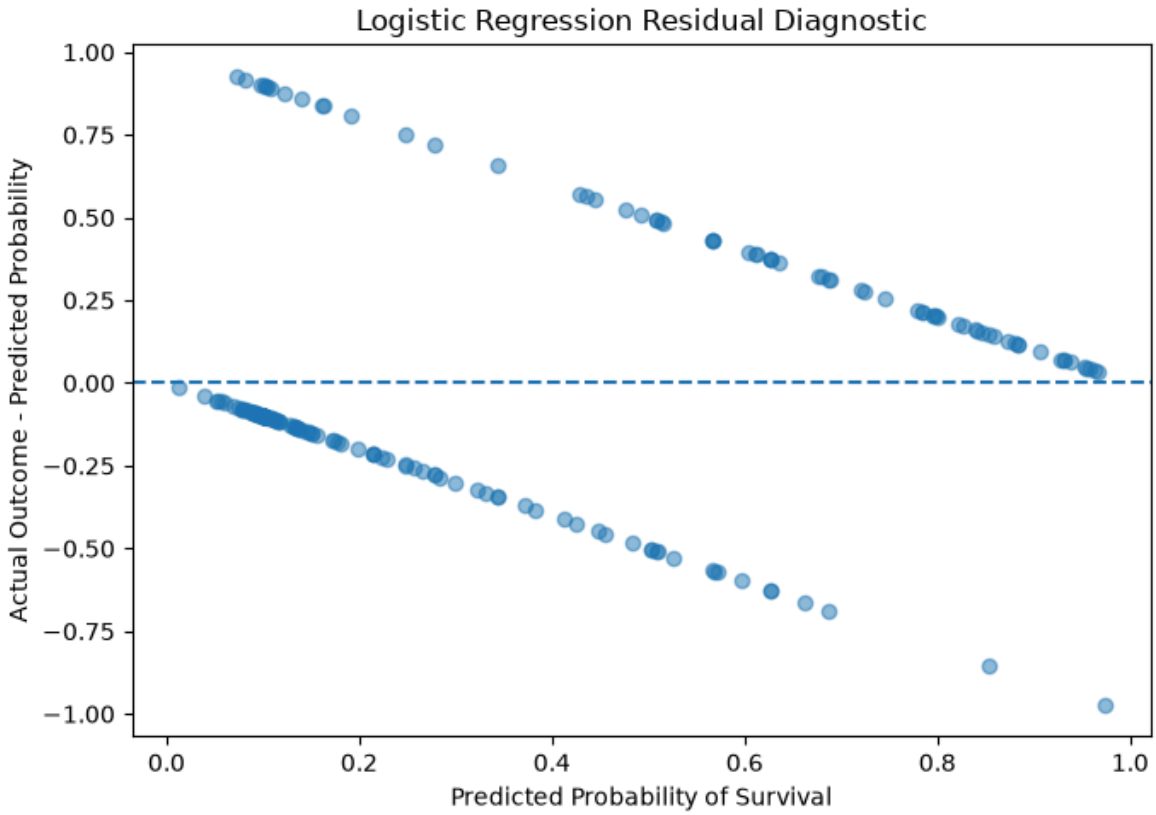
In [18]:
```python
cm = confusion_matrix(y_test, y_pred)

plt.figure(figsize=(6, 4))
sns.heatmap(
    cm, annot=True, fmt="d", cmap="Blues",
    xticklabels=["Predicted No", "Predicted Yes"],
    yticklabels=["Actual No", "Actual Yes"]
)
plt.title("Logistic Regression - Confusion Matrix")
plt.xlabel("Prediction")
plt.ylabel("Actual")
plt.tight_layout()
plt.show()
```



plt.figure(figsize=(7, 5)) plt.scatter(y_prob, y_test - y_prob, alpha=0.5) plt.axhline(0, linestyle="--") plt.xlabel("Predicted Probability of Survival") plt.ylabel("Actual Outcome - Predicted Probability") plt.title("Logistic Regression Residual Diagnostic") plt.tight_layout() plt.show()

In [20]:
```
plt.figure(figsize=(7, 5))
plt.scatter(y_prob, y_test - y_prob, alpha=0.5)
plt.axhline(0, linestyle="--")
plt.xlabel("Predicted Probability of Survival")
plt.ylabel("Actual Outcome - Predicted Probability")
plt.title("Logistic Regression Residual Diagnostic")
plt.tight_layout()
plt.show()
```



Week 3: Statistical Modeling and Hypothesis Testing

1. Research Question

Are passenger characteristics such as gender, age, passenger class, fare, and family size associated with survival on the Titanic?

2. Hypothesis Testing

A Chi-square test was conducted to examine the relationship between gender and survival.

- **Chi-square statistic:** 260.717
- **P-value:** 1.197×10^{-58}
- **Significance level:** 0.05

Since the p-value is less than 0.05, the null hypothesis was rejected. The results indicate a statistically significant association between gender and survival.

3. Logistic Regression Results

A Logistic Regression model was trained using passenger class, gender, age, fare, siblings/spouses, and parents/children aboard.

- **Accuracy:** 80.4%
- **ROC-AUC:** 0.852
- **Test samples:** 179

The accuracy indicates that the model correctly classified approximately 80.4% of passengers in the test set. The ROC-AUC score of 0.852 indicates good ability to distinguish between passengers who survived and those who did not.

4. Confusion Matrix Interpretation

	Predicted Not Survived	Predicted Survived
Actually Not Survived	95	15
Actually Survived	20	49

The model correctly identified 95 passengers who did not survive and 49 passengers who survived. It incorrectly classified 15 non-survivors as survivors and 20 survivors as non-survivors.

5. Classification Report

- **Class 0 (Not Survived):** Precision = 0.83, Recall = 0.86, F1-score = 0.84.
- **Class 1 (Survived):** Precision = 0.77, Recall = 0.71, F1-score = 0.74.

The model performed somewhat better at identifying passengers who did not survive than at identifying passengers who survived.

6. Limitations

The dataset contains missing values, and the model uses only a subset of available passenger characteristics. Other relevant factors may not be included. Statistical association does not prove causation, and the results should not automatically be generalized to other populations.

7. Conclusion

The Chi-square test found a statistically significant association between gender and survival. Logistic Regression achieved 80.4% accuracy and a ROC-AUC of 0.852 on the held-out test set. These results suggest that the selected passenger characteristics provide useful information for predicting Titanic survival.